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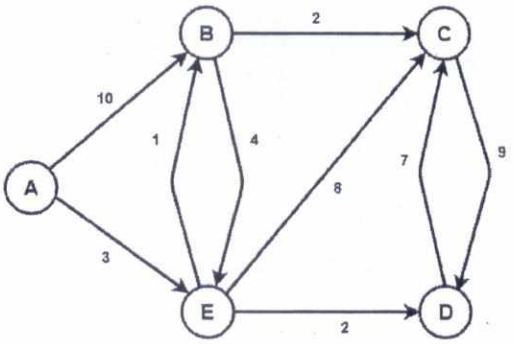
Thapar Institute of Engineering & Technology, Patiala

Department of Electrical and Instrumentation Engineering

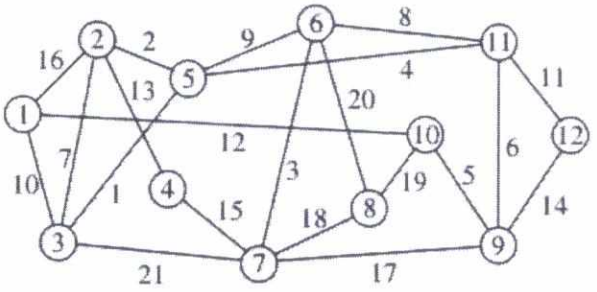
End Semester Test

B.E (BME)- 3rd Year	Code: UBM605
Dec 14, 2024	Course: Data Structures and Algorithms
M. Marks: 35	Name of Faculty: Dr. Alok Kumar Shukla

Note: Attempt Any Five questions in a proper sequence with justification. Assume missing data, if any suitably.

Q. 1(a)	Given the following sequence of letters and asterisks: EAS*Y*QUE***ST***IO*N*** Consider the stack data structure, supposing two operations, push and pop. Suppose that for the above sequence, each letter (such as E) corresponds to a push of that letter onto the stack, and each asterisk (*) corresponds to a pop operation on the stack. Show the sequence of values returned by the pop operations.	CO1, L1 (3.5+3.5)
(b)	Consider the following sequence of operations on an empty stack - push(54), push(52), pop(), push(55), push (62), m = pop() and the following sequence of operations on an empty queue - enqueue(21), enqueue(24), dequeueQ (), enqueue (28), enqueue(32), n = dequeueQ. What will be the output of m + n? During these operations, you need to show the content of the stack and queue.	
Q. 2 (a)	Write the pseudocode of Dijkstra's shortest path algorithm. In the figure given below, multiple shortest paths exist between Source ('A) and Destination ('D') vertices. Determine the shortest path and cost to all the vertices that can be reached from source vertex '**A**' using Dijkstra's shortest path algorithm. Illustrate each intermediate step.  	CO2, L2 (3.5+3.5)

b)	Explain a singly linked list. The **fun1 ()** function is supposed to reverse a singly linked list. Write one line that is missing at the end of the function.
	
	struct node
	{
	int data;
	struct node* next;
	};
	
	/ * head_refer is a double pointer that points to the head (or start) pointer of the linked list * /
	
	static void **fun1** (struct node** head_refer)
	{
	struct node* prev = NULL;
	struct node* current = *head_refer;
	struct node* next;
	while (current != NULL)
	{
	next = current->next;
	current->next = prev;
	prev = current;
	current = next;
	}	

	_____/* FILL STATEMENT HERE*/ }	
Q. 3 (a)	In an adjacency list representation of an undirected simple graph G=(V,E), each edge (u,v) has two adjacency list entries: [v] in the adjacency list of u, and [u] in the adjacency list of v. These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If |E|=m and |V|=n, and the memory size is not a constraint, now estimate the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list. Show all the needed steps.	CO3, L4 (3.5+3.5)
(b)	What are the different rotations present in the AVL tree? Create an AVL tree starting with an empty tree using the following values in the same order as given below: 20, 11, 5, 32, 40, 2,4, 27, 23, 28, 50. Also, Label each node in the intermediate trees with a balance factor.	
4.(a)	Construct max-heap on the array (**A**): *81, 89, 9, 11, 14, 76, 54, 22*. Show all intermediate steps involved in the construction of the max-heap. Write the algorithm of Max-Heapify on given array **A**. Discuss heap sort average-case complexity.	CO4, L2 (3.5+3.5)
(b)	Define the minimum spanning tree (MST) and enlist any two properties of MST for an undirected, weighted, connected graph G {V, E}. Consider a graph in the following figure. if we start with node 10 in V_T as the starting node and use Prim's algorithm to construct the minimum spanning tree, give the order in which nodes enter V_T. Also, estimate the minimum cost of the spanning tree.  	
Q. 5(a)	Construct a binary search tree (BST) by inserting the elements in the given order. 60, 25, 72, 15, 30, 68, 101, 13, 18, 47, 70, 34 I. Determine the height of the constructed binary search tree. II. Find the total number of nodes in the left subtree of the root. III. Perform the In-order, Pre-order, and post-order tree traversals and provide the output.	CO2, L4 (3.5+3.5)
(b)	A hash table size is given as 13. Double hashing is used to resolve collisions. Suppose the first hash function is H1(k) = k mod 13. The second hash function is H2(k) = 7 - (k mod 7). **Solve the following queries**. A. Resolve collisions by double hashing to show the possible hash table after inserting the following order keys (k):18, 41, 22, 44, 59, 32, 31, 73 B. 59 stored at which location in the hash table. C. Is the value 13 a good choice for table size? If no, give a valid reason and find the best table size value for the hash table.	
6.	Suppose there are seven cars parked in a parking area. A token number is given to the owner of each car, and the same number is stamped on the car by the parking caretaker. The parked cars have the following sequence of numbers 1,2,3,4,5,6,7 and are parked in the same order as previously mentioned. The caretaker reverses the order of a portion of the sequence of cars and arranges the vehicles in order 1,5,4,3,2,6,7. Write an efficient algorithm to find the order of cars after placing a portion of cars in the reverse order using a linked list.	CO3, L3 (7)

********** END**********